

Lagrangian Duality Framework

Primal: $\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{s.t. } g_i(\mathbf{x}) \leq 0, h_j(\mathbf{x}) = 0$



Lagrangian: $\mathcal{L}(\mathbf{x}, \mu, \lambda) = f(\mathbf{x}) + \sum_i \mu_i g_i(\mathbf{x}) + \sum_j \lambda_j h_j(\mathbf{x})$



Dual function: $D(\mu, \lambda) = \min_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \mu, \lambda)$



Dual problem: $\max_{\mu \geq 0, \lambda} D(\mu, \lambda)$



Weak duality: $d^* \leq p^*$ Strong duality: $d^* = p^*$ (convex + CQ)